

# INFOSYS SPRING BOARD

## VIRTUAL INTERNSHIP 6.0

**MILESTONE1 :**

**Orientation & Data Loading**

**Domain: Data Visualization**

**BATCH NO :13**

**PROJECT NAME:**

**HotelRevAI - AI Driven Revenue Analysis for  
Hotels**

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## INTRODUCTION:

In today's data-driven business environment, organizations rely heavily on analytics tools to understand performance, optimize operations, and support strategic decision-making. The hospitality industry, in particular, generates large volumes of data related to bookings, customer behavior, pricing, and revenue. However, raw hotel data is often unstructured, inconsistent, and difficult to analyze without proper preprocessing and analytical tools.

The project titled "AI Driven Revenue Analysis for Hotels" focuses on transforming raw hotel booking and sales data into meaningful insights using Power BI. The project emphasizes data cleaning, transformation, and feature engineering to ensure accuracy and reliability in analysis. Key challenges such as separated date fields, missing values, NA entries, and the absence of direct revenue metrics were addressed using Power Query.

By creating calculated columns such as Arrival Date, Total Nights, and Revenue, the dataset was enhanced to support time-based and performance analysis. Interactive dashboards were then developed to visualize revenue trends, booking patterns, and category-wise and region-wise performance. Although the project does not implement machine learning models directly, it follows an AI-driven analytical approach by enabling intelligent, data-backed insights through automated calculations and dynamic visualizations.

This project demonstrates how business intelligence tools like Power BI can be effectively used to support decision-making in the hotel industry, making it highly relevant for real-world applications in data analytics and business intelligence roles.

## Abstract:

The project “**AI Driven Revenue Analysis for Hotels**” aims to analyze hotel booking and sales data to extract meaningful revenue insights using Power BI. Real-world hotel datasets often contain missing values, inconsistent formats, and incomplete business metrics, making direct analysis difficult. In this project, raw data was cleaned and transformed using Power Query, and important calculated columns such as Arrival Date, Total Nights, and Revenue were created. Interactive dashboards were designed to visualize revenue trends, booking behavior, and performance across different categories and regions. The project demonstrates how AI-driven analytical thinking combined with business intelligence tools can support informed decision-making in the hospitality industry.

## Problem Statement”

The hospitality industry generates large amounts of booking and transactional data, but this data is often not analysis-ready. The available hotel dataset contained separated date components, missing and NA values, zero-value records, and lacked direct revenue and stay duration fields. These issues made it difficult to perform accurate revenue analysis and trend identification. The challenge was to clean and preprocess the dataset, derive meaningful business metrics, and present insights in an interactive and understandable manner using Power BI.

## Objectives:

- To preprocess and clean raw hotel and sales data
- To handle missing, NA, and zero-value records effectively
- To perform time-based and performance-based analysis
- To design interactive dashboards using Power BI
- To support AI-driven, data-based decision making in hotel revenue management.

## Dataset Description:

The project uses hotel booking and sales datasets in Excel format obtained from **kaggle** . The dataset contains information related to booking dates, stay duration, pricing, and sales performance. Key attributes include arrival year, arrival month, arrival day, average daily rate (ADR), number of week nights and weekend nights stayed, sales, profit, category, and region. The dataset required extensive cleaning and transformation before analysis.

## Tools and Technologies Used:(till now):

- **Power BI Desktop** – Data visualization and dashboard development
- **Power Query** – Data cleaning, transformation, and feature creation
- **Microsoft Excel** – Dataset storage and initial inspection

## Conclusion:

The project “**AI Driven Revenue Analysis for Hotels**” successfully demonstrates how raw hotel data can be transformed into actionable insights using Power BI. Through effective data cleaning, calculated feature

creation, and interactive visualization, the project highlights the importance of business intelligence in the hospitality sector. The skills and techniques applied in this project are directly applicable to real-world data analytics and business intelligence roles.